

Atlas: AI-Powered Customer Support Agent with Fine-Tuned Phi-4-mini and RAG Pipeline

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1. Introduction

- E-commerce platforms require scalable, intelligent customer support
- Rule-based chatbots fail on long-tail queries and nuanced language
- LLMs offer fluent responses but hallucinate without grounding
- Atlas combines fine-tuned SLM + RAG + deterministic tool routing
- Goal: accurate, empathetic, grounded support at small-model cost

2. Motivation & Problem Statement

- Support tickets cost ~\$5–15 each; automation saves 40–70 %
- Customers demand instant resolution of order, refund & policy queries
- Challenge: hallucination, identity verification, latency, cost
- Research question: can a fine-tuned 3.8 B SLM + RAG match GPT-4-class accuracy?

3. Dataset

- Amazon product catalog — 359 K items (title, category, price, description)
- Policy knowledge base — 45 hand-authored entries (returns, refunds, shipping)
- Fine-tune corpus — 2,400 synthetic customer-support Q&A pairs
- Evaluation set — 50 hand-crafted test cases across 8 intent categories

4. System Architecture

- FastAPI backend · PostgreSQL (orders, sessions) · Qdrant vector DB
- HuggingFace Inference Endpoint — fine-tuned Phi-4-mini adapter (A100 GPU)
- Retrieval: sentence-transformers/all-MiniLM-L6-v2 (384-dim), cosine top-6
- Deterministic tool router (keyword regex) — 8 tools
- lookup_order · process_refund · cancel_order · get_account_info · search_product_knowledge · search_policy_knowledge
- escalate_to_human · send_customer_email
- Identity verification gate — email/order ID required before PII access
- Two-layer safety net: stall detection + data-presence → template fallback
- Docker Compose stack: api · postgres · qdrant · mailhog

5. Model Selection & Fine-Tuning

- Evaluated 4 open-weight models on 20 qualitative prompts:
- LLaMA3.2-1B · SmolLM2.1.7B · Qwen2.5-1.5B · Phi-4-mini-instruct
- Selected: Phi-4-mini-instruct (3.8 B) — best instruction following & tone
- Fine-tune: QLoRA 4-bit (BF16 quantization) on Google Colab A100
- 2,400 Q&A pairs · 3 epochs · batch size 4 · learning rate 2e-4
- Adapter deployed to HuggingFace Hub; served via A100 Inference Endpoint
- Avg first-token latency: ~3–5 s (warm) · cold-start: 30–60 s

12. Future Work & Limitations

- Deploy to-RAG baseline for direct ablation comparison
- Add cross-encoder re-ranking over Qdrant top-k results
- Streaming responses (SSE) to reduce perceived latency
- Stronger identity: OTP or OAuth2 instead of email-only
- Multi-turn context compression for conversations > 10 turns
- Cancellation flow refinement — communicate post-delivery return path
- Mobile-responsive frontend & persistent session storage

References

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- Lewis et al. Retrieval-Augmented Generation for NLP. NeurIPS 2020.
- Li et al. G-Eval: NLG Eval via GPT-4. EMNLP 2023.
- Microsoft Phi-4-mini-instruct. HuggingFace Hub, 2024.
- Amazon Product Review Dataset 2023. Kaggle.
- sentence-transformers/all-MiniLM-L6-v2. HuggingFace, 2021.

6. Evaluation Framework

- 50 hand-crafted test cases — 8 intent categories:
- order lookup · refund · cancellation · product recommendation
- policy FAQ · escalation · account info · general greeting
- Metric:
- ROUGE-L — surface lexical overlap with reference answers
- Task Success Rate — binary: correct tool call + grounded response
- G-Eval (1–5) — Gemini 2.5 Flash Judge across 5 dimensions:
- Relevance · Faithfulness · Completeness · Tone & Empathy · Groundedness
- Identity-sensitive cases pre-seeded with customer_id to bypass ID gate

7. Overall Evaluation Results

Metric Score

ROUGE-L 0.191

Task Success Rate 72.3 %

G-Eval Average (1–5) 3.79

Relevance 4.24

Tone & Empathy 4.18

Faithfulness 3.75

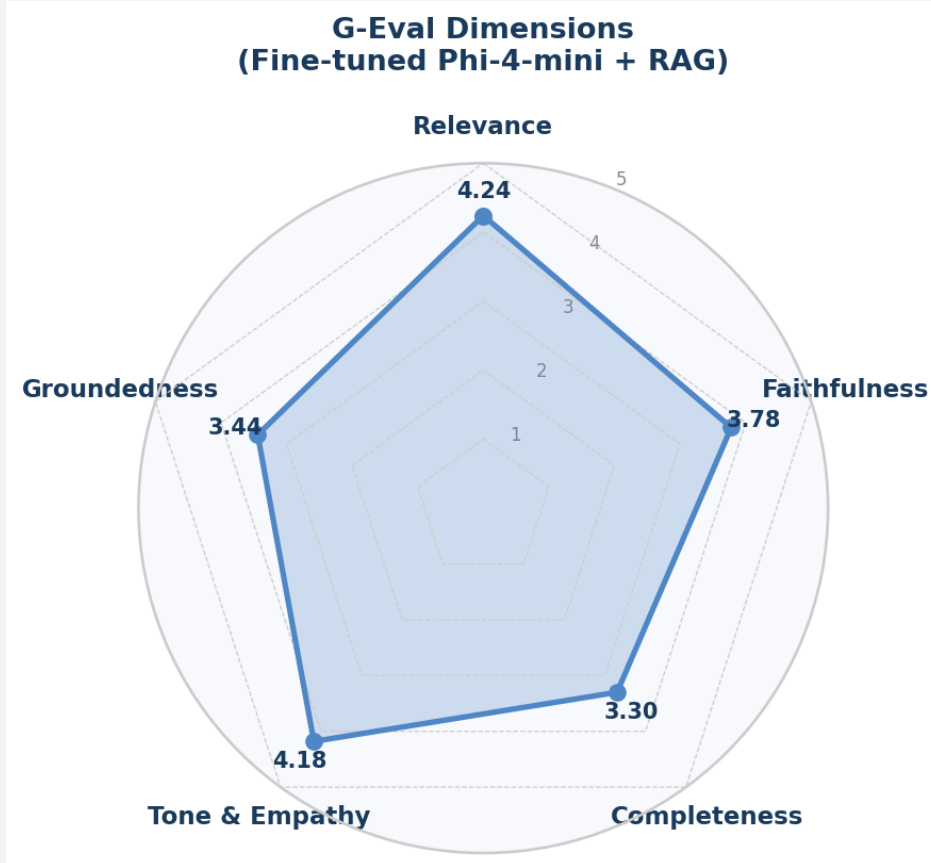
Groundedness 3.44

Completeness 3.30

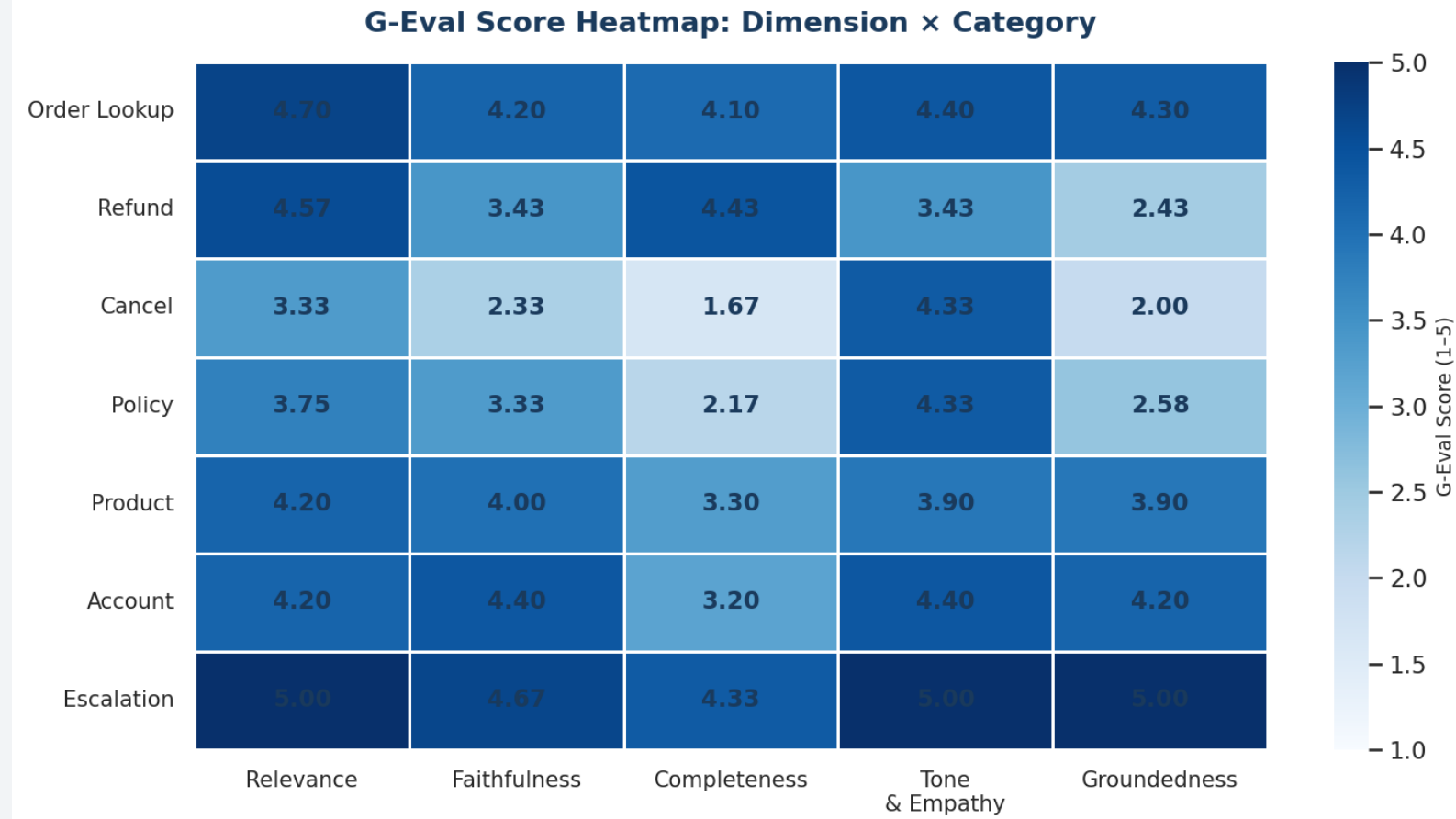
Best category: Escalation 4.80

Weakest category: Cancellation 2.73

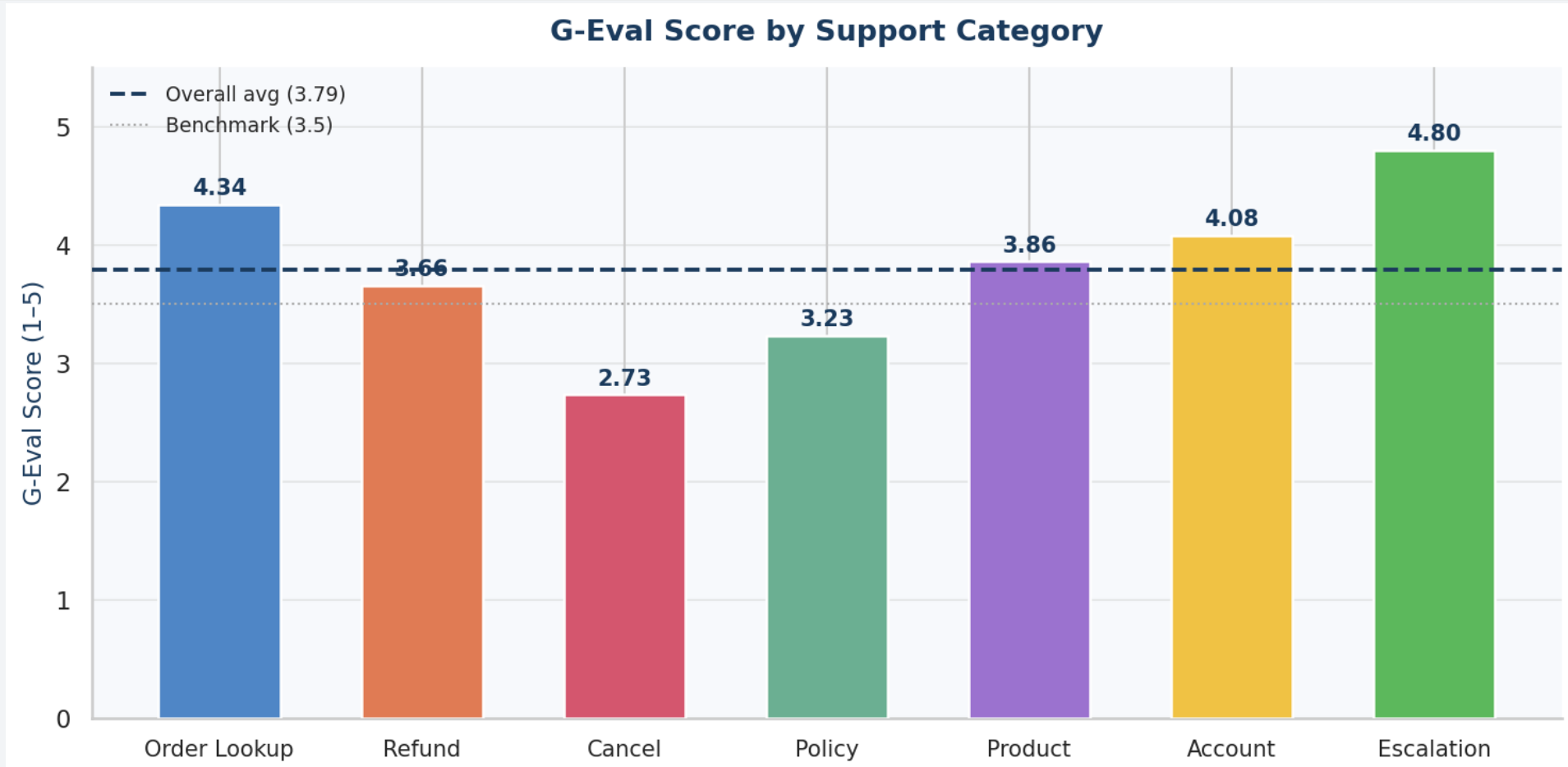
8. G-Eval Dimension Radar



9. G-Eval Heatmap: Dimension × Category



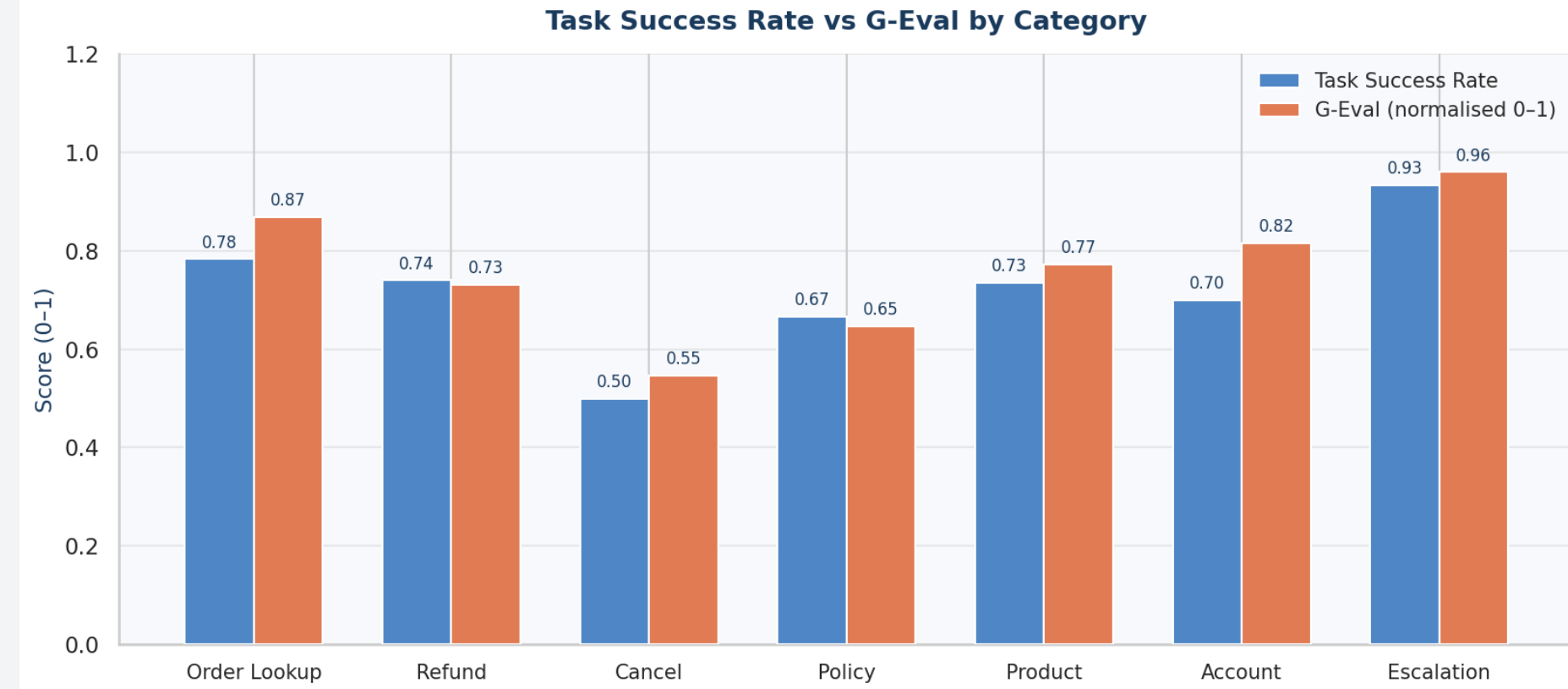
10. G-Eval Average by Intent Category



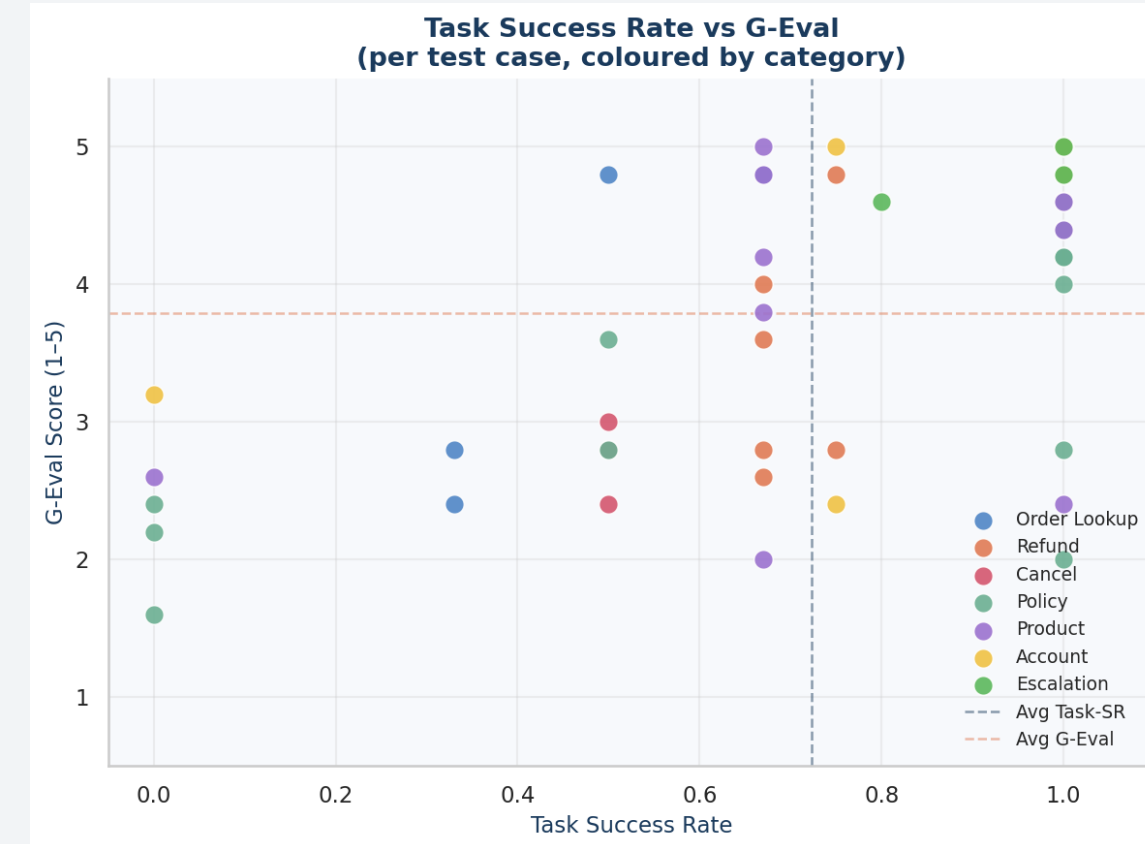
13. Conclusions

- Fine-tuned Phi-4-mini + RAG achieves 72.3 % task success on e-commerce support
- Deterministic routing + fallback templates are critical — ROUGE-L alone misleads
- Escalation best handled (4.80 G-Eval); cancellation most problematic (2.73)
- Identity gate prevents data leakage with minimal friction for legitimate users
- RAG grounding reduces hallucination; re-ranking would further improve precision

11. Task Success Rate & G-Eval by Category



12. Task Success vs G-Eval Per Test Case

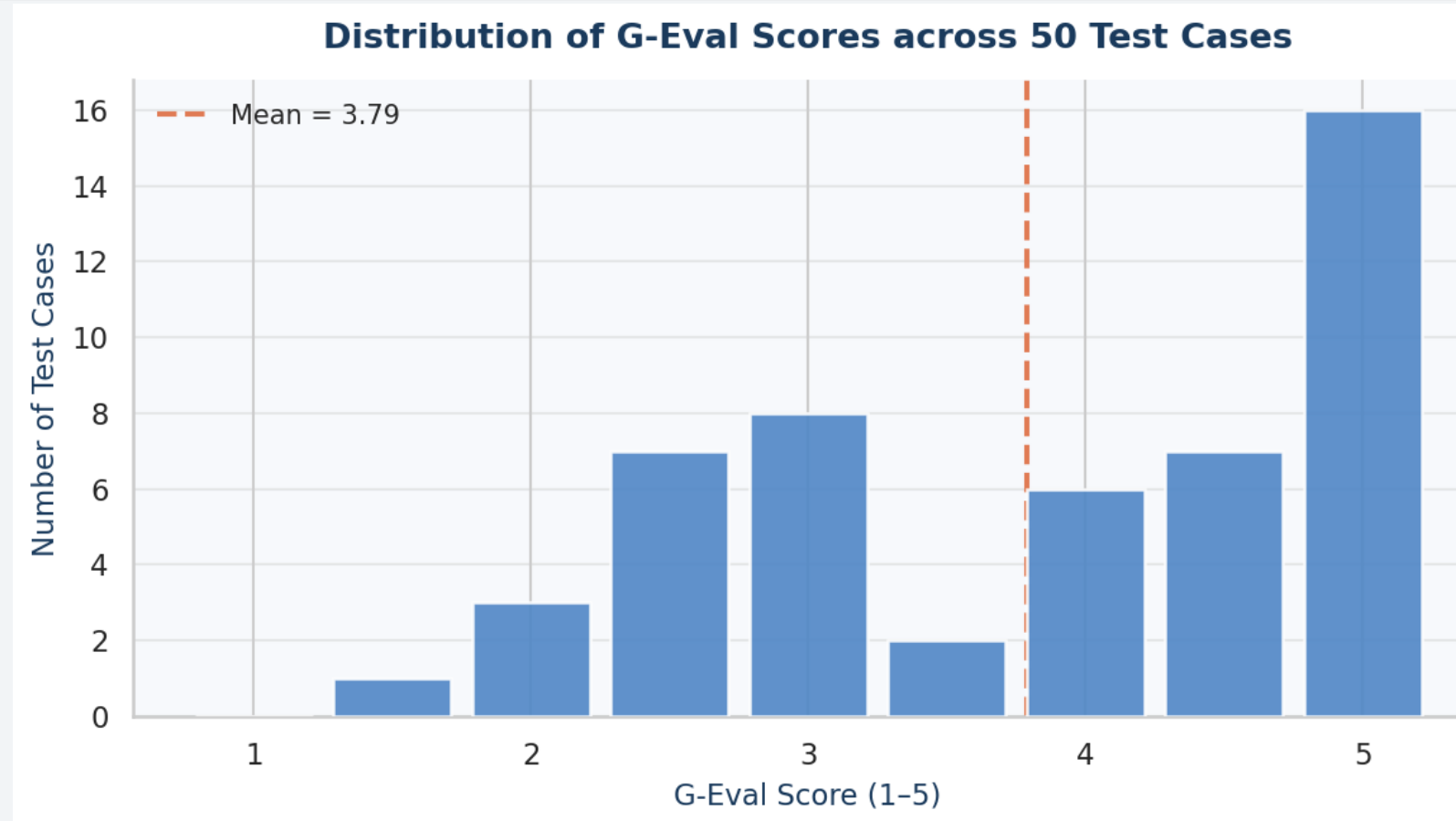


Summary Dashboard — All Metrics Overview

Atlas — Overall Evaluation Metrics (Fine-tuned Phi-4-mini + RAG · 50 test cases)



G-Eval Score Distribution



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